YANG–MILLS THEORY AND THE MASS GAP

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Abstract. For four-dimensional lattice Yang–Mills theory with a compact simple gauge group $G \subset \mathrm{U}(N)$ and Ricci curvature κ_G we prove the lattice mass gap $m \geq \operatorname{arccosh}\bigl(1 + \frac{1}{12}\bigl(\frac{\kappa_G}{N\beta} - \frac{29}{2}\bigr)\bigr)$ whenever $N\beta < \frac{2}{29}\kappa_G$. We prove that a spectral gap ϖ of the dynamics that resamples cubes of physical size ℓ, uniform in the lattice spacing, gives every nontrivial continuum limit the mass gap $\Delta \geq 16\varpi/(625e^2\ell)$, and that this cube gap holds at strong coupling and for massive free fields; along the weak-coupling tuning that the continuum problem requires it remains open.

1. Introduction

The Yang–Mills existence and mass gap problem asks for a quantum Yang–Mills theory on $\mathbb{R}^4$ with a compact simple gauge group that satisfies the axioms of constructive quantum field theory, and for a number $\Delta > 0$ such that the spectrum of the Hamiltonian lies in $\{0\} \cup [\Delta, \infty)$ [19]. In the Euclidean formulation the theory is a probability measure on gauge fields whose Schwinger functions satisfy the Osterwalder–Schrader axioms [26, 27]; the mass gap is then exponential decay of the truncated two-point functions of gauge-invariant observables at the rate Δ. The natural regularisation is Wilson's lattice gauge theory [33]: an element Q_e of $G \subset \mathrm{U}(N)$ sits on every link e of the lattice $a\mathbb{Z}^4$, and a configuration has the weight $\exp(N\beta \sum_p \operatorname{Re}\operatorname{Tr} Q_p)$, where Q_p is the ordered product around the plaquette p.[1] Asymptotic freedom forces the continuum limit $a \to 0$ to be taken with $\beta \to \infty$, that is, at weak coupling.

At small β the lattice theory has a unique state, exponential clustering and a transfer-matrix gap, by convergent expansions [28, 30]; Bałaban's renormalisation group [7, 8, 9], as presented by Dimock [13], gives ultraviolet stability in finite volume at weak coupling, and Magnen, Rivasseau and Sénéor constructed the theory with an infrared cutoff [24]. No existing result gives a mass gap in physical units. This paper proves three things about this problem, and states precisely what is missing.

1.1. The lattice gap with explicit constants.

Let G be a compact connected Lie group with semisimple Lie algebra, identified with its image under a faithful unitary representation in $\mathrm{U}(N)$, and let $\kappa_G > 0$ be its least Ricci curvature for the bi-invariant metric $\langle X, Y\rangle = -\operatorname{Tr}(XY)$. For a simple group $\kappa_G = h^\vee/(2\ell_\rho)$, with $h^\vee$ the dual Coxeter number and ℓ_ρ the Dynkin index of the representation; for $\mathrm{SU}(N)$, $\kappa_G = N/2$.

Theorem A. *Suppose $N|\beta| < \frac{2}{29}\kappa_G$. For every gauge-invariant local observable the expectations in boxes with free boundary converge, along every exhausting sequence of boxes, to a state ω that is invariant under the lattice symmetries; truncated correlations of Lipschitz*

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[1] This is the normalisation of [29, 2]. The conventional inverse coupling is $\beta_W = N^2\beta$, and for SU(2), identified with the unit quaternions, $\beta_W = 4\beta$.

observables decay at every rate below

$$m_G(\beta) = \operatorname{arccosh}\Big(1 + \frac{1}{12}\Big(\frac{\kappa_G}{N|\beta|} - \frac{29}{2}\Big)\Big) > 0,$$

and for $\beta \geq 0$ *the state is reflection positive and its transfer matrix has spectral gap* $m \geq m_G(\beta)$. *For* $G = \mathrm{SU}(2)$ *the same holds for* $|\beta_W| < \frac{1}{3}$ *with* $m \geq -\log(3\beta_W)$.

Two ingredients carry the proof. The first is a gauge: a field of straight lines on $\mathbb{Z}^4$ that contains exactly one link of every plaquette (Sec. 3). Setting these links to the identity loses no gauge-invariant information, and every remaining link then interacts with exactly twelve others instead of eighteen. The second is a Hessian bound in this gauge, $|\operatorname{Hess} S| \leq \frac{29}{2}$ for the action $S = \sum_p \operatorname{Re}\operatorname{Tr} Q_p$, proved by a divergence identity and a positivity statement for one plaquette (Sec. 5); with the Ricci curvature of G it gives a Bakry–Émery bound [5, 6] with an explicit decay rate. Without the gauge the sharp Hessian bound is 16 [2], so the gauge enlarges the range of every group by the factor 32/29 and supplies the rate. The SU(2) statement combines the gauge with the sharp Kantorovich modulus $\frac{1}{4}$ of the von Mises–Fisher family [2] in Dobrushin's criterion [14, 15].

1.2. **A continuum mass gap from one cube.** The lattice gap of Theorem A is measured in lattice units and vanishes in physical units as $a \to 0$. The second result isolates a single estimate, posed on one scale, that controls the gap in physical units. On a periodic lattice let $\mathcal{P}_x$ resample the links of the cube of side n at x from their conditional law under the Wilson measure μ. The condition $\mathrm{BP}(n, \varpi)$ asks that

$$\varpi \operatorname{Var}_\mu(f) \leq \frac{1}{n^4} \sum_x \mu((f - \mathcal{P}_x f)^2)$$

for every gauge-invariant f: the dynamics that resamples cubes, each link at rate one, has spectral gap at least ϖ on gauge-invariant functions. The condition involves no gauge fixing and no supremum over boundary conditions.

Theorem B. *Let* $a_k \to 0$ *and* $\beta_k \geq 0$ *be couplings, and suppose that there are* $\varpi > 0$, $\ell > 0$ *and integers* $n_k \geq 2$ *with* $n_k a_k \leq \ell$ *such that* $\mathrm{BP}(n_k, \varpi)$ *holds for the Wilson measure at* β_k *on all large tori. Then every continuum limit of the corresponding lattice theories has Hamiltonian* H *with* $\sigma(H) \subset \{0\} \cup [\Delta, \infty)$,

$$\Delta = \frac{16\varpi}{625 e^2 \ell},$$

and 0 *is a simple eigenvalue; if the limit is nontrivial,* H *has a mass gap of at least* Δ.

The proof (Sec. 6) turns the spectral gap into decay of covariances through a finite-speed bound for the cube dynamics, at a rate of order $1/n$ in lattice units, hence of order $1/\ell$ in physical units, and passes this decay through the Osterwalder–Schrader reconstruction.

1.3. **Where the cube condition holds.**

Theorem C. (i) *At* $\beta = 0$, $\mathrm{BP}(n, 1)$ *holds for every* n. *For* $24N|\beta| < \kappa_G$ *and* n *large,* $\mathrm{BP}(n, 1 - \alpha_n)$ *holds with* $\alpha_n = N|\beta|(18n - 3)/((\kappa_G - 24N|\beta|)n^2) \to 0$; *for* SU(2) *the range is* $|\beta_W| < \frac{2}{9}$.

(ii) *For the free scalar field of lattice mass* μ *in* d *dimensions the best constant is* $\varpi = \frac{\mu^2}{n^d}\langle \mathbf{1}, (-\Delta_D + \mu^2)^{-1}\mathbf{1}\rangle$, *with* $-\Delta_D$ *the Dirichlet Laplacian of the cube, and with* $s = n\mu$ *(the mass times the physical size of the cube)*

$$\Big(\frac{5}{6}\Big)^d \frac{s^2}{s^2 + 10d} \leq \varpi \leq \frac{s^2}{s^2 + 16d/9}.$$

Part (ii) is the model case of the continuum tuning: the constant is uniform in the lattice spacing at fixed $m\ell$, and it vanishes as $m \to 0$. The cube condition is therefore a quantitative statement of mass generation at the scale ℓ.

1.4. **What remains.** Theorem A is a strong-coupling theorem. The Clay problem needs the opposite end of the coupling range: the continuum limit lives at $\beta \to \infty$, where the range $N\beta < \frac{2}{29}\kappa_G$ of Theorem A and the range of Theorem C(i) do not reach. Two statements would complete the proof of a continuum mass gap, and neither is proved here:

(D) $\mathrm{BP}(n_k, \varpi)$ with $n_k a_k \le \ell$ along the asymptotic-freedom tuning $\beta_k \to \infty$, uniformly in k;

(E) convergence of the renormalised Schwinger functions to a nontrivial limit satisfying the Osterwalder–Schrader axioms.

Given (D) and (E), Theorem B gives the mass gap $\Delta \ge 16\varpi/(625e^2\ell)$. Section 9 explains why (D) is the heart of the matter: by Theorem C(ii) it cannot hold without a dynamically generated mass, it fails at fixed cube size at weak coupling, and it fails in the Coulomb phase of the abelian theory, so its proof has to use the non-abelian structure of G at every scale up to ℓ.

1.5. **Related work.** Strong-coupling clustering goes back to Osterwalder and Seiler [28] and Seiler [30]; probabilistic treatments include the survey of Chatterjee [11], the Langevin approach of Shen, Zhu and Zhu [29], whose Bakry–Émery argument we sharpen, the area law in enlarged regions of Cao, Nissim and Sheffield [10], and correlation decay for finite gauge groups at weak coupling by Adhikari and Cao [1]. In two dimensions the continuum measure is known [18, 16, 23]. The Kantorovich modulus of the SU(2) single-link law and the sharp Hessian bound without gauge fixing are proved in [2], and an earlier and much longer account of the one-cube reduction is [3].

I. J. Sessford [31] has formalised in Lean 4 with Mathlib, without unproved statements and with only the standard axioms, several ingredients of the strong-coupling analysis on periodic lattices: the Haar probability on $\mathrm{SU}(N)$, an infinite-volume Gibbs state obtained as a limit of torus states, site reflection positivity and the transfer matrix, the Holley–Stroock comparison, Dobrushin's covariance bounds, and the passage from decay of correlations to a transfer-matrix gap (one direction of Proposition 2.1). For the periodic state it proves a transfer-matrix gap for every N at $\beta_W < 5.9 \cdot 10^{-5}$, and explicit single-link heat-bath gaps on boxes of side fifteen for SU(2) at $\beta_W \le 0.166$ and for SU(3) at $\beta_W \le 0.38$ (his coupling is $\beta_W/2$). It also proves, by a Poincaré inequality and a light-cone argument different from Lemma 6.2, that an ungauged one-cube influence condition gives a lattice gap of order $1/n$; this is a variant of the lattice step, Theorem 6.3 and Corollary 6.4, at a smaller rate. The continuum step, the gauge fixing and the curvature bounds are not part of that development, which shares with this paper the open hypothesis (D).

2. Lattice gauge theory and the transfer matrix

Write $e_0, \dots, e_3$ for the unit vectors of $\mathbb{Z}^4$, (x, μ) for the link from x to $x + e_\mu$, and $Q_{(x,\mu)}^{-1}$ for the variable of the reversed link. A box Λ is a product of four finite intervals, with its set $E(\Lambda)$ of links and its set $P(\Lambda)$ of plaquettes. For a plaquette at x in the plane $\mu < \nu$ with links $e_1 = (x, \mu)$, $e_2 = (x + e_\mu, \nu)$, $e_3 = (x + e_\nu, \mu)$, $e_4 = (x, \nu)$, put $Q_p = Q_{e_1} Q_{e_2} Q_{e_3}^* Q_{e_4}^*$. With dQ the Haar probability measure of G, the Wilson measure of a box with free boundary is

$$W_\Lambda(dQ) = \frac{1}{Z_\Lambda} \exp\Big(N\beta \sum_{p \in P(\Lambda)} \operatorname{Re} \operatorname{Tr} Q_p\Big) \prod_{e \in E(\Lambda)} dQ_e,$$

and the Wilson measure μ_L of the torus $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$ is defined in the same way with all links and plaquettes of Λ_L. Both are invariant under the gauge transformations $Q_{(x,y)} \mapsto g_x Q_{(x,y)} g_y^{-1}$. A function of finitely many links is a local observable, and it is gauge invariant if it is invariant under these transformations. For $\beta \ge 0$ the measures are reflection positive for reflections in lattice hyperplanes through sites and in hyperplanes bisecting links, on boxes symmetric under the reflection and on tori of even side [28].

Let ω be a state on the gauge-invariant local observables of $\mathbb{Z}^4$ that is invariant under translations and reflection positive in this sense. The Osterwalder–Schrader construction [26, 27, 28] gives a Hilbert space $\mathcal{H}$ with a unit vector Ω, a map $F \mapsto \Psi_F$ from the observables at nonnegative times to $\mathcal{H}$ with dense range, and a self-adjoint transfer matrix $0 \le \mathbb{T} \le 1$ with $\mathbb{T}\Omega = \Omega$ and

$$\langle \Psi_F, \mathbb{T}^t \Psi_H \rangle = \omega\big(\vartheta(\bar{F})\, \tau_t H\big), \qquad t \in \mathbb{Z}_{\ge 0},$$

where ϑ is the time reflection and τ_t the time translation; Ψ_F is the class of F for the positive semidefinite form $\omega(\vartheta(\bar{F})H)$, and $\|\Psi_F\|^2 \le \omega(|F|^2)$. The *spectral gap* of $\mathbb{T}$ is m if its spectrum on $\Omega^\perp$ lies in $[0, e^{-m}]$.

Proposition 2.1. *Let $m > 0$ and let $\mathcal{F}$ be a class of gauge-invariant local observables, supported at nonnegative times, that is dense in $L^2(\omega)$ on each bounded time support. If for every $F \in \mathcal{F}$ there is $C_F < \infty$ with*

$$\omega(\vartheta(\bar{F})\tau_t F) - |\omega(F)|^2 \le C_F e^{-mt} \qquad (t \ge 0),$$

then $\mathbb{T}$ has spectral gap at least m.

Proof. Put $\Psi'_F = \Psi_F - \langle \Omega, \Psi_F \rangle \Omega$, so that the left side equals $\langle \Psi'_F, \mathbb{T}^t \Psi'_F \rangle = \int \lambda^t \, d\sigma_F(\lambda)$, with σ_F the spectral measure of $\mathbb{T}$ at Ψ'_F. For $\delta > 0$ this is at least $\sigma_F((e^{-m+\delta}, 1])e^{(-m+\delta)t}$, which exceeds $C_F e^{-mt}$ for large t unless $\sigma_F((e^{-m+\delta}, 1]) = 0$. So the spectral projection of $(e^{-m}, 1]$ annihilates every Ψ'_F. These vectors are dense in $\Omega^\perp$, because $\|\Psi_F\|^2 \le \omega(|F|^2)$ and the class is dense, so the projection vanishes on $\Omega^\perp$. □

The decay of a single two-point function does not suffice: it places only one spectral measure in $[0, e^{-m}]$, and the density of the class is what gives the gap.

3. The line field

A *direction field* is a map $\sigma : \mathbb{Z}^4 \to \{0, 1, 2, 3\}$; its links are $F = \{(x, \sigma(x)) : x \in \mathbb{Z}^4\}$. At each site put $\pi_x(\mu) = \sigma(x + e_\mu)$. The plaquette at x in the plane $\{\mu, \nu\}$ contains

$$[\sigma(x) \in \{\mu, \nu\}] + [\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu] \tag{3.1}$$

links of F.

Lemma 3.1. *Every plaquette contains exactly one link of F if and only if, at every site x, π_x fixes $\sigma(x)$ and permutes the other three directions cyclically.*

Proof. Fix x, let $s = \sigma(x)$ and let R be the other three directions. For the plane $\{s, \nu\}$ the count (3.1) is $1 + [\pi_x(s) = \nu] + [\pi_x(\nu) = s]$, which equals one for all $\nu \in R$ exactly when π_x fixes s and maps R into R. For a plane $\{\mu, \nu\} \subset R$ the count is $[\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu]$. Draw an arrow $\mu \to \pi_x(\mu)$ whenever $\pi_x(\mu) \in R \setminus \{\mu\}$; each vertex has at most one outgoing arrow, and the condition asks for exactly one arrow on each of the three pairs. Three arrows on three vertices with out-degree at most one give out-degree one everywhere, and such a tournament is a directed three-cycle. □

From now on σ is the field of period two of Table 2 in Appendix A; in each row π_x fixes $\sigma(x)$ and is a three-cycle on the other directions, so every plaquette contains exactly one link of F.[2] Call the links outside F *active* and write A for their set.

Lemma 3.2.

(i) *F is a disjoint union of bi-infinite straight lines, one through every site; hence $F \cap E(\Lambda)$ is a forest for every box Λ.*

(ii) *An active link $e = (x, \mu)$ lies in six plaquettes. In four of them the link of F shares an end point with e; in the other two it is opposite to e.*

[2] A field with one link of F at every site and one in every plaquette can exist only in four dimensions: a site carries $d(d-1)/2$ plaquettes at which it is the base point and a link lies in $2(d-1)$ plaquettes, and the two counts agree only for $d = 4$.

(a) the line field in the slice $x_3 = 0$ (b) one plaquette in the gauge

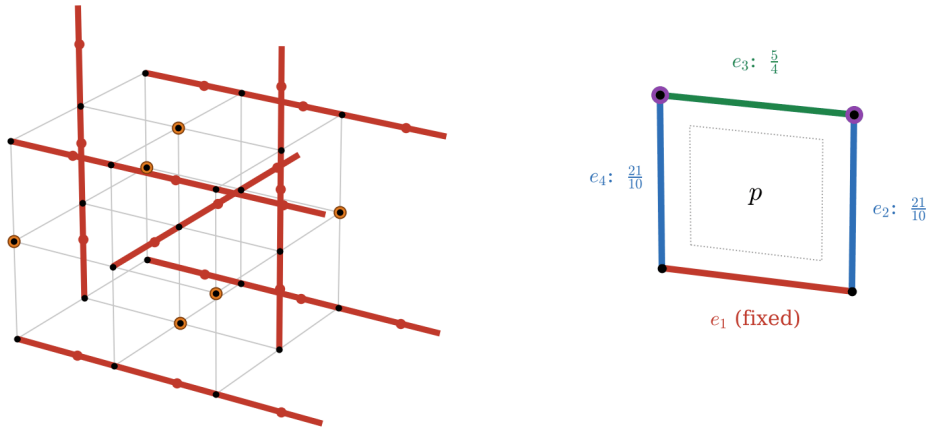


FIGURE 1. (a) The line field in the slice $x_3 = 0$ of a box of side three; thick segments are links of F, rings mark sites whose line leaves the slice in direction 3. (b) One plaquette with its fixed link e_1 and the weights $\frac{21}{10}$, $\frac{5}{4}$ of Lemma 5.2 on its active links.

(iii) *Every active link shares a plaquette with exactly twelve other active links, two in each of its plaquettes, and two links share at most one plaquette.*

Proof. (i) Since π_x fixes $\sigma(x)$, $\sigma(x + e_{\sigma(x)}) = \sigma(x)$, and since σ has period two, also $\sigma(x - e_{\sigma(x)}) = \sigma(x)$. If $(x - e_\nu, \nu) \in F$, straightness at $x - e_\nu$ gives $\sigma(x) = \nu$, so the incoming link of F at x is unique and continues the outgoing one. A line meets a box in a segment.

(ii) A link of F sharing an end point with e is the outgoing or incoming link of F at x, of direction $s = \sigma(x) \neq \mu$, or at $x + e_\mu$, of direction $t = \pi_x(\mu) \in R \setminus \{\mu\}$. They lie in the two plaquettes of e in the plane $\{\mu, s\}$ and the two in the plane $\{\mu, t\}$, four distinct plaquettes as $s \neq t$. In the two plaquettes of the remaining plane $\{\mu, \rho\}$ the unique link of F touches neither end of e, so it is the opposite link.

(iii) Each plaquette of e contains e, one link of F and two other links. □

4. Gauge fixing and the free-boundary state

For a function O of the links let O^F be O with every link of F set to the identity.

Lemma 4.1 (gauge fixing)**.** *Let μ_Λ^F be the probability measure on $G^{A \cap E(\Lambda)}$ with density proportional to $\exp(N\beta \sum_{p \in P(\Lambda)} \operatorname{Re} \operatorname{Tr} Q_p^F)$. For every gauge-invariant O on $G^{E(\Lambda)}$, $W_\Lambda(O) = \mu_\Lambda^F(O^F)$.*

Proof. Each segment of $F \cap E(\Lambda)$ is a path $x_0, \dots, x_k$. Given the variables on F, put $g_{x_0} = 1$ and $g_{x_{i+1}} = g_{x_i} Q_{(x_i, x_{i+1})}$ along every segment. The transformed field equals the identity on $F \cap E(\Lambda)$, and for fixed values on F the map from the active variables to the transformed ones is a product of translations, which preserve Haar measure. Gauge invariance of the action and of O gives the claim. □

No holonomy remains because $F \cap E(\Lambda)$ is a forest. The measures μ_Λ^F are finite-volume versions of the specification γ^F on G^A in which γ_e^F resamples an active Q_e from the law with density proportional to $\exp(N\beta \sum_{p \ni e} \operatorname{Re} \operatorname{Tr} Q_p^F)$ given the other active links. Fix a bi-invariant metric δ on G of diameter D and define the Kantorovich coefficients

$$C_{ee'} = \sup \frac{W_1(\gamma_e^F(\cdot \mid Q), \gamma_e^F(\cdot \mid Q'))}{\delta(Q_{e'}, Q'_{e'})},$$

the supremum over configurations that differ only at e'; W_1 is the Kantorovich distance [32], and $C_{ee'} = 0$ unless e and e' share a plaquette.

Lemma 4.2 (comparison)**.** *Suppose every row and column of $(C_{ee'})$ sums to at most $\alpha < 1$, and let $\lambda > 0$ with $\theta = \alpha e^{\lambda} < 1$. Let $S \subset A$ and let μ_1, μ_2 be probability measures on G^A invariant under γ_e^F for $e \notin S$. Let $T \subset A \setminus S$ be finite, $w_b = e^{-\lambda \operatorname{dist}(b,T)}$ with the distance of link midpoints, and suppose $\sum_{z\in S} w_z < \infty$. There is a coupling of μ_1 and μ_2 with $\varrho(b) = \mathbb{E}\,\delta(\sigma_b, \sigma'_b)$ satisfying*

$$\sum_{b \notin S} w_b \varrho(b) \le \frac{\theta D}{1-\theta} \sum_{z \in S} w_z,$$

and $\mu_1 = \mu_2$ if $S = \emptyset$.

Proof. The couplings form a weakly compact set on which $\Psi(\pi) = \sum_b w_b \mathbb{E}_\pi \delta(\sigma_b, \sigma'_b)$ is continuous; let π^* minimise it. For $e \notin S$ replace (σ_e, σ'_e) by a sample of an optimal coupling of $\gamma_e^F(\cdot \mid \sigma)$ and $\gamma_e^F(\cdot \mid \sigma')$, chosen measurably [32, Cor. 5.22]. Both μ_i are invariant, so this is again a coupling, and changing the links that enter γ_e^F one at a time gives $\varrho(e) \le \sum_{e'} C_{ee'} \varrho(e')$ by minimality. Multiply by w_e and sum over $e \notin S$; where $C_{ee'} \neq 0$, $w_e \le e^{\lambda} w_{e'}$, so $\sum_{e \notin S} w_e \varrho(e) \le \theta[\sum_{e' \notin S} w_{e'} \varrho(e') + D \sum_{z \in S} w_z]$. If $S = \emptyset$ this forces $\varrho \equiv 0$. □

Conditioning on the links of the support of H [22, Ch. 3] and applying the lemma to the two conditional measures gives, for the Gibbs measure μ of γ^F, a local F' that is L-Lipschitz in each link and a bounded H with finite support disjoint from that of F',

$$|\operatorname{Cov}_\mu(F', H)| \le \|H\|_\infty L \,\frac{\theta D}{1-\theta}\, |\operatorname{supp} H|\, e^{-\lambda \operatorname{dist}(\operatorname{supp} F', \operatorname{supp} H)}. \tag{4.1}$$

Proposition 4.3. *Suppose every row and column of $(C_{ee'})$ sums to at most $\alpha < 1$. Then γ^F has a unique Gibbs measure μ_F; for every gauge-invariant local O the limit $\omega(O) = \lim W_\Lambda(O)$ exists along every sequence of boxes exhausting $\mathbb{Z}^4$ and equals $\mu_F(O^F)$; ω is invariant under translations and the hypercubic group; truncated correlations of Lipschitz gauge-invariant observables decay at every rate below $-\log\alpha$; and for $\beta \ge 0$, ω is reflection positive and its transfer matrix has spectral gap at least $-\log\alpha$.*

Proof. Gibbs measures exist by compactness and coincide by Lemma 4.2 with $S = \emptyset$. Extend μ_Λ^F by Haar measure outside Λ and let S be the set of active links with a plaquette outside $P(\Lambda)$; off S the extended measure is invariant under γ_e^F. Lemma 4.2 with T the support of O^F bounds $|\mu_\Lambda^F(O^F) - \mu_F(O^F)|$ by $C_T R^3 e^{-\lambda R}$, R the distance from T to the complement of Λ, for Lipschitz O, and uniform approximation handles continuous O; Lemma 4.1 identifies $\mu_\Lambda^F(O^F)$ with $W_\Lambda(O)$. Since $W_\Lambda(\tau_v O) = W_{\Lambda - v}(O)$ and the limit does not depend on the exhausting sequence, ω is invariant; the line field is a device for computing W_Λ and is never moved. Reflection positivity of W_Λ on symmetric boxes passes to the limit. The decay is (4.1), since $\operatorname{Cov}_\omega(O, O') = \operatorname{Cov}_{\mu_F}(O^F, O'^F)$. Lipschitz gauge-invariant functions are dense in the gauge-invariant part of $L^2(\omega)$ on every support, being Lipschitz functions on the compact quotient by the gauge group, so Proposition 2.1 gives the gap. □

Proof of Theorem A for SU(2). Identify SU(2) with the unit quaternions and use the chordal metric. For an active link e and a plaquette $p \ni e$ let s_p be the product of the other three variables, ordered so that $\operatorname{Re}(Q_e s_p) = \frac{1}{2}\operatorname{Tr} Q_p$, with links of F equal to one. Since $N\beta \operatorname{Re}\operatorname{Tr} Q_p = 4\beta \langle Q_e, \bar{s}_p \rangle$ in $\mathbb{R}^4$, the conditional law of Q_e is the von Mises–Fisher law $\nu_b(dx) \propto e^{\langle x, b\rangle} dx$ on S^3 with $b = 4\beta \sum_{p \ni e} \bar{s}_p$. Moving a partner e' by δ moves one staple by an isometric image of that displacement, hence b by $4|\beta|\delta$, and the sharp quarter of [2], $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b - b'|$, gives $C_{ee'} \le |\beta|$. By Lemma 3.2(iii) every row and column has twelve nonzero entries, so $\alpha \le 12|\beta| = 3|\beta_W|$, and Proposition 4.3 applies for $|\beta_W| < \frac{1}{3}$. □

The value $\frac{1}{3}$ is the end of this argument: at $b = 0$ the derivative of ν_b has Kantorovich norm exactly $\frac{1}{4}$, the six staples of an active link can be arranged with sum zero while one of

them moves freely, and so every row sum of the matrix is exactly $3|\beta_W|$. Without the gauge every link has eighteen partners and the same argument stops at $\beta_W = \frac{2}{9}$ [2, Prop. 13].

5. Curvature in the gauge

In this section $G \subset \mathrm{U}(N)$ is closed and connected with Lie algebra $\mathfrak{g}$ and the inner product $\langle X, Y\rangle = -\operatorname{Tr}(XY)$. A tangent vector at a field Q is $v = (X_e Q_e)$ with $X_e \in \mathfrak{g}$, and $|v|^2 = \sum_e |X_e|^2$. For the plaquette p at x in the plane $\mu < \nu$, with links $e_1, \dots, e_4$ as in Sec. 2, transport the vectors to x,

$$Y_1 = X_{e_1}, \quad Y_2 = Q_{e_1} X_{e_2} Q_{e_1}^*, \quad Y_3 = Q_{e_4} X_{e_3} Q_{e_4}^*, \quad Y_4 = X_{e_4},$$

and write $W = Q_p$. Along the geodesics $Q_e(t) = e^{tX_e} Q_e$ the plaquette variable is $Q_p(t) = e^{tY_1} e^{tY_2} W e^{-tY_3} e^{-tY_4}$, and differentiating twice gives the second derivative of $\operatorname{Re}\operatorname{Tr} Q_p$,

(5.1)
$$f_p = \sum_{i=1}^{4} \operatorname{Re}\operatorname{Tr}(Y_i^2 W) + 2\operatorname{Re}\operatorname{Tr}(Y_1 Y_2 W) + 2\operatorname{Re}\operatorname{Tr}(Y_3 Y_4 W) - 2 \sum_{i\in\{1,2\},\, j\in\{3,4\}} \operatorname{Re}\operatorname{Tr}(Y_i W Y_j),$$

so that $\operatorname{Hess} S_0(v,v) = \sum_p f_p$ for $S_0 = \sum_p \operatorname{Re}\operatorname{Tr} Q_p$. The *corner form* of p is

$$c_p = 2\langle Y_1, Y_4\rangle - 2\langle Y_1, Y_2\rangle + 2\langle Y_2, W Y_3 W^*\rangle - 2\langle Y_3, Y_4\rangle; \tag{5.2}$$

it collects the inner products of the vectors of two links that meet at a right angle at a corner of p.

Lemma 5.1 (divergence identity)**.** *On $\mathbb{Z}^4$, or on a box with free boundary, for every field Q and every square-summable X,*

$$\sum_y \Big[|D_y X|^2 + \sum_\mu \big| X_{y,\mu} + Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu} \big|^2 \Big] = 4|X|^2 + \sum_p c_p,$$

where $D_y X = \sum_\mu [X_{y,\mu} - Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu}]$ is the covariant divergence and terms with a link outside the box are omitted.

Proof. At a site y let o_μ be the vector of the outgoing link of direction μ and i_μ the transported vector of the incoming one, so that $D_y X = \sum_\mu (o_\mu - i_\mu)$. Every link is outgoing at one end and incoming at the other, so the squares $|o_\mu|^2 + |i_\mu|^2$, which occur once in $|D_y X|^2$ and once in $|o_\mu + i_\mu|^2$, add up to $4|X|^2$. The collinear cross term $-2\langle o_\mu, i_\mu\rangle$ of $|D_y X|^2$ cancels the term $+2\langle o_\mu, i_\mu\rangle$ of $|o_\mu + i_\mu|^2$. Every remaining cross term of $|D_y X|^2$ pairs two perpendicular ends at y, and such a pair is a corner of exactly one plaquette, with the sign and transport of (5.2); in a box two perpendicular ends at y span a plaquette of the box. □

Lemma 5.2 (one plaquette)**.** *Let $k \in \{1,2,3,4\}$, let $k \pm 1$ (mod 4) be the positions adjacent to k and $k+2$ the opposite one. For every N, every $W \in \mathrm{U}(N)$ and all $Y_1, \dots, Y_4 \in \mathfrak{u}(N)$ with $Y_k = 0$,*

$$\pm f_p + \tfrac{9}{10} c_p \le \tfrac{21}{10}\big(|Y_{k-1}|^2 + |Y_{k+1}|^2\big) + \tfrac{5}{4}|Y_{k+2}|^2.$$

Proof. Replacing W by $-W$ changes the sign of f_p and fixes c_p, so the sign + suffices, and conjugation by a unitary matrix preserves both sides, so we may take $W = \operatorname{diag}(w_a)$. The difference $\mathcal{F}$ of the two sides then splits into a block for each diagonal position a, in the real variables $y_i = -i(Y_i)_{aa}$, and a block for each pair $a \ne b$, in the complex variables $z_i = (Y_i)_{ab}$, since $(Y_i)_{ba} = -\bar{z}_i$. Appendix B shows that every block is positive definite. □

Theorem 5.3 (Hessian bound)**.** *For every closed subgroup $G \subset \mathrm{U}(N)$, every box or $\mathbb{Z}^4$, every field Q and every tangent vector v that vanishes on the links of F,*

$$|\operatorname{Hess} S_0(v,v)| \le \tfrac{29}{2}|v|^2.$$

TABLE 1. The curvature $\kappa_G = h^\vee/(2\ell_\rho)$ in a faithful representation of least dimension N, and the range of Theorem A for $\beta_W = N^2\beta$.

Group	ρ	$h^\vee$	ℓ_ρ	κ_G	$\beta_W <$
$\mathrm{SU}(N)$, $N \geq 2$	$\mathbb{C}^N$	N	1	$\frac{N}{2}$	$\frac{N^2}{29}$
$\mathrm{SO}(N)$, $N \geq 5$	$\mathbb{R}^N$	$N-2$	2	$\frac{N-2}{4}$	$\frac{N(N-2)}{58}$
$\mathrm{Sp}(n)$, $n \geq 2$	$\mathbb{C}^{2n}$	$n+1$	1	$\frac{n+1}{2}$	$\frac{2n(n+1)}{29}$
G_2	7	4	2	1	$\frac{14}{29}$
F_4	26	9	6	$\frac{3}{4}$	$\frac{39}{29}$
E_6	27	12	6	1	$\frac{54}{29}$
E_7	56	18	12	$\frac{3}{4}$	$\frac{84}{29}$
E_8	248	30	60	$\frac{1}{4}$	$\frac{124}{29}$

Proof. Every plaquette has exactly one link of F, on which X vanishes, and by Lemma 3.2(ii) each active link is adjacent to that link in four of its plaquettes and opposite to it in two. Summing Lemma 5.2 over the plaquettes,

$$\sum_p (\pm f_p + \tfrac{9}{10} c_p) \leq (4 \cdot \tfrac{21}{10} + 2 \cdot \tfrac{5}{4})|X|^2 = \tfrac{109}{10}|X|^2.$$

By Lemma 5.1, $\frac{9}{10}\sum_p c_p = \frac{9}{10}\Sigma - \frac{36}{10}|X|^2$ with $\Sigma \geq 0$ the left side of that lemma. Hence $\pm \operatorname{Hess} S_0(v,v) \leq (\frac{109}{10} + \frac{36}{10})|X|^2 = \frac{29}{2}|X|^2$; links near the boundary of a box lie in fewer plaquettes and receive less. Both lemmas concern vectors in $\mathfrak{u}(N)$, so the bound holds on every closed subgroup, whose geodesics $e^{tX}Q_e$ with $X \in \mathfrak{g}$ are geodesics of $\mathrm{U}(N)$. □

The factor $\frac{9}{10}$ lets the divergence identity pay for the corners at the price $\frac{36}{10}$ while the blocks of one plaquette stay positive. Without the gauge the sharp bound is 16 [2].

Lemma 5.4 (Ricci curvature)**.** *Let $G \subset \mathrm{U}(N)$ be closed and connected with simple Lie algebra, and let $\mathfrak{t} \subset \mathfrak{g}$ be a maximal abelian subalgebra. Write the eigenvalues of* $\operatorname{ad} H$ *on $\mathfrak{g} \otimes \mathbb{C}$ and of H on $\mathbb{C}^N$, $H \in \mathfrak{t}$, as $i\alpha(H)$ and $i\mu(H)$, over the roots α and the weights μ counted with multiplicity. Then* $\mathrm{Ric}_G = \kappa_G\langle\cdot,\cdot\rangle$ *with*

$$\kappa_G = \frac{\sum_\alpha \alpha(H)^2}{4\sum_\mu \mu(H)^2} = \frac{h^\vee}{2\ell_\rho} \qquad (0 \neq H \in \mathfrak{t}).$$

Proof. For a bi-invariant metric $R(X,Y)Z = -\frac{1}{4}[[X,Y],Z]$ on left-invariant fields [25, Sect. 7], so for an orthonormal basis (E_i), $\mathrm{Ric}(X,X) = \frac{1}{4}\sum_i |[X,E_i]|^2 = -\frac{1}{4}\operatorname{Tr}(\operatorname{ad} X)^2$. The Killing form and $\langle\cdot,\cdot\rangle$ are invariant and $\mathfrak{g}$ is simple, so the Killing form is $-4\kappa\langle\cdot,\cdot\rangle$ for a constant κ, and on $\mathfrak{t}$ the two forms are $-\sum_\alpha \alpha(H)^2$ and $\sum_\mu \mu(H)^2$. For H the coroot of the highest root, $\sum_\alpha \alpha(H)^2 = 4h^\vee$ and $\sum_\mu \mu(H)^2 = 2\ell_\rho$ [21, Ch. 6]. □

For two links $e \neq e'$ of a plaquette p write $\operatorname{Hess}\operatorname{Re}\operatorname{Tr} Q_p(v,v) = \sum_{e,e' \in p} H^p_{ee'}(X_e, X_{e'})$ with $H^p_{ee'}(a,b) = H^p_{e'e}(b,a)$. By (5.1) a mixed block is, after isometric transport, $\pm\operatorname{Re}\operatorname{Tr}(abW)$ or $\pm\operatorname{Re}\operatorname{Tr}(aWb)$, so

$$|H^p_{ee'}(a,b)| \leq |a|\,|b|. \tag{5.3}$$

5.1. **Decay with an explicit rate.** Let A' be a finite set of active links and P' a finite set of plaquettes whose active links lie in A'. Call $S = \sum_{p \in P'} c_p \operatorname{Re}\operatorname{Tr} Q_p^F$, $0 \leq c_p \leq 1$, *admissible* if $|\operatorname{Hess} S(v,v)| \leq \frac{29}{2}|v|^2$ for v tangent to $G^{A'}$. By Theorem 5.3 the action $S_\Lambda = \sum_{p \in P(\Lambda)} \operatorname{Re}\operatorname{Tr} Q_p^F$ of a box is admissible, and so is every convex combination of admissible actions; by (5.3) and Lemma 3.2(iii) the mixed blocks of $\operatorname{Hess} S$ have norm at most one and each link has at most twelve partners. Let μ have density proportional to $e^{N\beta S}$ on $G^{A'}$, let $L = \Delta + N\beta\langle\nabla S, \nabla\cdot\rangle$ be its Langevin generator, self-adjoint in $L^2(\mu)$, and

$P_t = e^{tL}$. For weights $w > 0$ on A' put $\Gamma^w(f) = \sum_e w_e |\nabla_e f|^2$, with ∇_e the gradient in the factor of e; Γ is the case $w \equiv 1$.

Lemma 5.5. *Let G have semisimple Lie algebra, $\lambda \geq 0$, $w_e = e^{\lambda d_e}$ with $|d_e - d_{e'}| \leq 1$ for partners, and*

$$K_\lambda = \kappa_G - N|\beta|\Big(\tfrac{29}{2} + 12\big(\cosh\tfrac{\lambda}{2} - 1\big)\Big).$$

Then $\Gamma^w(P_t f) \leq e^{-2K_\lambda t} P_t \Gamma^w(f)$ for smooth f and $t \geq 0$.

Proof. Put $V = -N\beta S$. On the product $G^{A'}$ the projection to one factor is parallel and the Ricci tensor is block diagonal, so Bochner's formula applied factor by factor gives

$$\tfrac{1}{2} L\Gamma^w(g) - \Gamma^w(g, Lg) = \sum_e w_e\big(|\nabla\nabla_e g|^2 + \mathrm{Ric}_G(\nabla_e g, \nabla_e g)\big) + \sum_{e,e'} w_e \operatorname{Hess} V_{ee'}(\nabla_e g, \nabla_{e'} g).$$

With $u_e = \nabla_e g$, $v_e = \sqrt{w_e} u_e$ and the symmetry of $\operatorname{Hess} V$, the last sum is

$$\operatorname{Hess} V(v, v) + \sum_{e \neq e'} \Big(\cosh\tfrac{\lambda(d_e - d_{e'})}{2} - 1\Big) \operatorname{Hess} V_{ee'}(v_e, v_{e'}),$$

since $(w_e + w_{e'})/(2\sqrt{w_e w_{e'}}) = \cosh\frac{\lambda(d_e - d_{e'})}{2}$. The first term is at least $-\frac{29}{2} N|\beta||v|^2$; in the second only partners contribute, each with coefficient at most $\cosh\frac{\lambda}{2} - 1$ and a block of norm at most $N|\beta|$, and with twelve partners it is at least $-12(\cosh\frac{\lambda}{2} - 1)N|\beta||v|^2$. As $\mathrm{Ric}_G \geq \kappa_G$ and $|v|^2 = \Gamma^w(g)$, the left side is at least $K_\lambda \Gamma^w(g)$. For $\varphi(s) = P_s \Gamma^w(P_{t-s} f)$ this gives $\varphi' \geq 2K_\lambda \varphi$, hence $\varphi(0) \leq e^{-2K_\lambda t}\varphi(t)$. □

Lemma 5.6. *Suppose $K_\lambda > 0$. Let f, g be smooth functions of the links of A_f and B_g, and let D be the graph distance from A_f to B_g in the graph on A' whose edges join partners. Then*

$$|\operatorname{Cov}_\mu(f, g)| \leq \frac{e^{-\lambda D/2}}{K_\lambda} \|\nabla f\|_\infty \sum_{e \in B_g} \|\nabla_e g\|_\infty.$$

Proof. Lemma 5.5 with $\lambda = 0$ shows that $P_t f \to \mu(f)$ uniformly, so $\operatorname{Cov}_\mu(f, g) = \int_0^\infty \mu(\Gamma(P_t f, g))\, dt$. Let d_e be the smaller of D and the graph distance from e to A_f; then $w = 1$ on A_f, $w = e^{\lambda D}$ on B_g, and $\Gamma^w(f) \leq \|\nabla f\|_\infty^2$. For $e \in B_g$, Lemma 5.5 gives $|\nabla_e P_t f|^2 \leq e^{-\lambda D} e^{-2K_\lambda t} \|\nabla f\|_\infty^2$, and integration in t gives the bound. □

Proof of Theorem A. Let λ^* solve $K_{\lambda^*} = 0$, so that $\lambda^*/2 = m_G(\beta)$, and let $0 < \lambda < \lambda^*$. Lemma 5.6 with $S = S_\Lambda$ gives $|\operatorname{Cov}_{\mu_\Lambda^F}(f, g)| \leq C_{f,g} e^{-\lambda D/2}$ uniformly in the box. For boxes $\Lambda \subset \Lambda'$ interpolate $S_s = (1 - s)S_\Lambda + sS_{\Lambda'}$, which is admissible; for smooth f of the active links of Λ,

$$\frac{d}{ds}\mu_s(f) = N\beta \sum_{p \in P(\Lambda') \setminus P(\Lambda)} \operatorname{Cov}_{\mu_s}(f, \operatorname{Re}\operatorname{Tr} Q_p^F),$$

with $|\nabla_e \operatorname{Re}\operatorname{Tr} Q_p| \leq \sqrt{N}$. A plaquette outside $P(\Lambda)$ is at graph distance at least $R - 1$ from the support of f, R the distance to the complement of Λ, and at most $C(k+1)^4$ plaquettes lie at graph distance k, so $|\mu_{\Lambda'}^F(f) - \mu_\Lambda^F(f)| \leq C_f R^4 e^{-\lambda R/2}$. The limit exists along every increasing sequence of boxes, two sequences are compared through their intersections, and continuous gauge-invariant observables are uniform limits of smooth ones $e^{\tau\Delta}O$, the heat semigroup commuting with translations of G. With Lemma 4.1 the limit ω exists; invariance and reflection positivity follow as in Proposition 4.3. Since $\mathrm{Ric}_G \geq 0$, $e^{\tau\Delta}O$ has the Lipschitz constant of O [6, Sect. 3.3], so the decay passes to Lipschitz observables; two links of a plaquette have times differing by at most one, so observables at time separation t are at graph distance at least $t - c$, and truncated correlations decay at every rate below $m_G(\beta)$. Proposition 2.1 gives the gap. □

The rate behaves like $\log\frac{\kappa_G}{6N|\beta|}$ as $\beta \to 0$. The proof also gives a Poincaré inequality with constant $1/(\kappa_G - \frac{29}{2}N|\beta|)$ for every μ_Λ^F, uniformly in the box [6, Cor. 5.7.2].

6. One cube and the continuum gap

Setting. Let $G \subset \mathrm{U}(N)$ be compact and nontrivial and φ a positive continuous class function on G, such as $e^{N\beta \operatorname{Re}\operatorname{Tr}(\cdot)}$. On the torus $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$, L even and $L \geq 2n+4$, with links E and plaquettes P, let

$$\mu(dU) = \frac{1}{Z} \prod_{p \in P} \varphi(U_p) \prod_{e \in E} dU_e.$$

A link (y, ν) has base point y; differences of base points are taken in $(-L/2, L/2]^4$ and $|\cdot|_\infty$ is the corresponding distance. For $x \in \Lambda_L$ let V_x be the set of links with base point in $x + [0,n)^4$; every link lies in exactly n^4 of the L^4 cubes V_x. For $V \subset E$ let ∂V be the links outside V that share a plaquette with a link of V; the conditional law $\gamma^0_V(\cdot \mid U)$ of U_V given the other links has density proportional to $\prod_{p \cap V \neq \emptyset} \varphi(U_p)$ and depends only on $U_{\partial V}$. Let $\mathcal{P}_x f(U) = \int f \, d\gamma^0_{V_x}(\cdot \mid U)$, the conditional expectation given the links outside V_x, an orthogonal projection on $L^2(\mu)$, and let $\mathcal{A}$ be the closed subspace of gauge-invariant functions. By gauge covariance of $\gamma^0_{V_x}$, $\mathcal{P}_x$ maps $\mathcal{A}$ into itself.

Definition 6.1. For $n \geq 2$ and $\varpi > 0$, the condition $\mathrm{BP}(n, \varpi)$ for μ is

$$\varpi \operatorname{Var}_\mu(f) \leq \frac{1}{n^4} \sum_{x \in \Lambda_L} \mu\big((f - \mathcal{P}_x f)^2\big) \qquad \text{for every real } f \in \mathcal{A}.$$

Put $q_n = (2+1/n)^4$, so that $16 < q_n \leq \frac{625}{16}$ for $n \geq 2$, and let $\mathcal{K} = L^{-4} \sum_x \mathcal{P}_x$ be the chain that resamples one uniformly chosen cube; for $Y \subset \Lambda_L$ let $\mathcal{K}_Y = L^{-4}(\sum_{x \in Y} \mathcal{P}_x + (L^4 - |Y|)I)$ be the same chain with the moves outside Y suppressed. Write $y \rightsquigarrow z$ if $\partial V_z \cap V_y \neq \emptyset$. A link of ∂V_z has its base point in $z + [-1, n]^4$, so $y \rightsquigarrow z$ implies $|z - y|_\infty \leq n$, and for each z at most $(2n+1)^4 = q_n n^4$ points y have $y \rightsquigarrow z$. For finite $S \subset E$ let $Y_1(S) = \{y : V_y \cap S \neq \emptyset\}$ and $Y_{j+1}(S) = Y_j(S) \cup \{y : y \rightsquigarrow z \text{ for some } z \in Y_j(S)\}$, $Y_0(S) = \emptyset$.

Lemma 6.2 (finite speed)**.** *Let f be a bounded function of the links of a finite set S, let $m_0 \geq 0$ and $Y = Y_{m_0}(S)$. Then $\mathcal{K}^t_Y f$ depends only on the links of $S \cup \bigcup_{y \in Y} \partial V_y$, and for every integer $t \geq 0$*

$$\|\mathcal{K}^t f - \mathcal{K}^t_Y f\|_\infty \leq \frac{2|S|}{q_n} \|f\|_\infty \sum_{m > m_0} \frac{(q_n \tau)^m}{m!}, \qquad \tau = \frac{t n^4}{L^4}.$$

Proof. If h depends only on the links of W, then $\mathcal{P}_x h = h$ when $V_x \cap W = \emptyset$, and otherwise $\mathcal{P}_x h$ depends only on the links of $(W \setminus V_x) \cup \partial V_x$; induction gives the first claim. For the second run two chains from the same configuration with the same uniformly chosen cubes $x_1, x_2, \ldots$. If $x_s \notin Y$ the first chain resamples V_{x_s} and the second does nothing; if $x_s \in Y$ and the two configurations agree on ∂V_{x_s}, both receive one common sample, and otherwise they resample independently. The marginals are the chains $\mathcal{K}$ and $\mathcal{K}_Y$. A link where the configurations differ at time t lies in $V_{x_{s_m}}$ for some times $s_1 < \cdots < s_m \leq t$ with $x_{s_1} \notin Y$ and $x_{s_l} \rightsquigarrow x_{s_{l+1}}$, by induction on t. If the set of such links meets S, then $x_{s_m} \in Y_1(S)$ and $x_{s_{m-l}} \in Y_{l+1}(S)$, so $m > m_0$. For fixed m there are at most $t^m/m!$ choices of times, $n^4|S|$ of x_{s_m} and $q_n n^4$ of each earlier cube, each sequence having probability L^{-4m}; the difference of f is at most $2\|f\|_\infty$. □

For finite $S, S' \subset E$ let $d(S, S')$ be the least $|b - b'|_\infty$ over base points b of links of S and b' of links of S'.

Theorem 6.3 (decay from one cube)**.** *Suppose* $\mathrm{BP}(n, \varpi)$ *holds for μ. Then $\varpi \leq 4$, and for bounded real $f, g \in \mathcal{A}$ depending on the links of finite sets S_f, S_g with $d = d(S_f, S_g) \geq 3n-2$,*

$$|\operatorname{Cov}_\mu(f, g)| \leq \Big(e + \frac{k|S_f||S_g|}{64e^2}\Big) \|f\|_\infty \|g\|_\infty \, e^{-\varpi k/(e^2 q_n)}, \qquad k = \Big\lfloor \frac{d+2}{n} \Big\rfloor - 2.$$

In particular $|\operatorname{Cov}_\mu(f,g)| \le C(1+d)e^{-\varpi d/(e^2 q_n n)}$ *for every* d*, with* C *depending only on* $|S_f|, |S_g|, \|f\|_\infty, \|g\|_\infty$.

Proof. The bound $\varpi \le 4$. Take $f = \operatorname{Re}\operatorname{Tr} U_p$ for one plaquette. Each $I - \mathcal{P}_x$ is an orthogonal projection annihilating constants, so $\mu((f-\mathcal{P}_x f)^2) \le \operatorname{Var}_\mu(f)$, and it vanishes unless V_x contains a link of p, which happens for at most $4n^4$ points x.

Spectral part. On the functions of $\mathcal{A}$ with mean zero, $\mathcal{K}$ is self-adjoint with $0 \le \mathcal{K} \le I$, and $\mathrm{BP}(n,\varpi)$ reads $\langle h, (I-\mathcal{K})h\rangle \ge \epsilon\|h\|^2$ with $\epsilon = \varpi n^4/L^4$. So $\|\mathcal{K}^T h\|_2 \le e^{-\epsilon T}\|h\|_2$, and from $f - \mu(f) = \mathcal{K}^T(f-\mu(f)) + \sum_{t<T}\mathcal{K}^t(I-\mathcal{K})f$,

$$\operatorname{Cov}_\mu(f,g) = \langle \mathcal{K}^T(f-\mu f), g-\mu g\rangle + \sum_{t=0}^{T-1}\langle \mathcal{K}^t f, (I-\mathcal{K})g\rangle,$$

whose first term is at most $e^{-\epsilon T}\|f\|_\infty\|g\|_\infty$.

Local part. Let $Z_g = Y_1(S_g)$, so $|Z_g| \le n^4|S_g|$ and $(I-\mathcal{K})g = L^{-4}\sum_{x\in Z_g}(g - \mathcal{P}_x g)$; if h does not depend on the links of V_x then $\langle h, g - \mathcal{P}_x g\rangle = 0$. Put $m_0 = k-1$ and $Y = Y_{m_0}(S_f)$. A link of V_x, $x \in Z_g$, lying in S_f or in ∂V_y with $y \in Y$ would join base points of S_f and S_g at distance at most $n(m_0+3) - 3 \le n\lfloor (d+2)/n\rfloor - 3 < d$, which is impossible. So $\mathcal{K}_Y^t f$ does not depend on the links of V_x, $x \in Z_g$, and by Lemma 6.2

$$\big|\langle \mathcal{K}^t f, (I-\mathcal{K})g\rangle\big| = \big|\langle \mathcal{K}^t f - \mathcal{K}_Y^t f, (I-\mathcal{K})g\rangle\big| \le \frac{4|S_f||S_g|n^4}{q_n L^4}\|f\|_\infty\|g\|_\infty \sum_{m\ge k}\frac{(q_n\tau_t)^m}{m!}.$$

Summing over $t < T$ with $\tau = Tn^4/L^4$, the second term is at most

$$\frac{4|S_f||S_g|\tau}{q_n}\,\|f\|_\infty\|g\|_\infty \sum_{m\ge k}\frac{(q_n\tau)^m}{m!}.$$

Choice of T. Take $T = \lfloor kL^4/(e^2 q_n n^4)\rfloor$, so that $z = q_n\tau \le k/e^2$ and $\epsilon T \ge \varpi k/(e^2 q_n) - 1$, using $\varpi \le 4$ and $L^4 \ge 4n^4$. Since $k! \ge (k/e)^k$, $\sum_{m\ge k} z^m/m! \le (ez/k)^k e^z \le e^{-k(1-e^{-2})}$, and $\tau/q_n \le k/(256e^2)$. As $\varpi/(e^2 q_n) < 1 - e^{-2}$, both terms are bounded as stated. For the second form use $k > d/n - 3$, and for $d < 3n-2$ the trivial bound $\|f\|_\infty\|g\|_\infty$. □

A *torus limit* at β is a state ω on the gauge-invariant local functions of $\mathbb{Z}^4$ with $\omega(O) = \lim_j \mu_{L_j}(O)$ along some $L_j \to \infty$; by compactness every sequence has a subsequence along which one exists.

Corollary 6.4 (lattice gap)**.** *Let* $\beta \ge 0$ *and suppose* $\mathrm{BP}(n,\varpi)$ *holds for* μ_L *for all* L *in an infinite set of even integers* $L \ge 2n+4$*. Then every torus limit along this set is invariant under translations, reflection positive, and its transfer matrix has spectral gap at least*

$$m_n = \frac{\varpi}{e^2 q_n n} \ge \frac{16\varpi}{625 e^2 n}.$$

Proof. Translation invariance and reflection positivity are identities and inequalities among finitely many local functions of μ_L [28], and pass to ω. For a continuous gauge-invariant F at positive times the supports of $\vartheta(\bar F)$ and $\tau_t F$ are at distance d with $|d - t| \le c_F$, and Theorem 6.3 on Λ_L gives $|\operatorname{Cov}_{\mu_L}(\vartheta(\bar F), \tau_t F)| \le C_F(1+t)e^{-m_n t}$ uniformly in L. In the limit $|\omega(\vartheta(\bar F)\tau_t F) - |\omega(F)|^2| \le C_F(1+t)e^{-m_n t}$, and Proposition 2.1, applied with every $m < m_n$, gives the gap. □

6.1. **The continuum limit.** Let $a_k \to 0$ be lattice spacings with $a_k = 2^{-k}a_0$, and β_k couplings. Let ω_k be lattice states with transfer matrices $\mathbb{T}_k$ on spaces $\mathcal{H}_k$ with vacua Ω_k. Let $\mathcal{T}$ be a complex vector space of test functions on $\mathbb{R}^4$ supported in $\{t > 0\}$, and for each k let $f \mapsto O_k(f)$ be a linear map to gauge-invariant local observables at positive times; the example is $O_k(f) = Z_k \sum_x a_k^4 f(a_k x)\Phi(x)$ with a local field Φ and a renormalisation constant Z_k.

Definition 6.5. A *continuum limit* of (ω_k, O_k) is a Hilbert space $\mathcal{H}$ with a unit vector Ω, a self-adjoint $H \geq 0$ with $H\Omega = 0$, and a linear map $\psi : \mathcal{T} \to \mathcal{H}$ with dense range such that for all $f, g \in \mathcal{T}$ and dyadic $\tau \geq 0$

$$\langle \psi(f), e^{-\tau H}\psi(g)\rangle = \lim_k \omega_k\big(\vartheta(\overline{O_k(f)})\,\tau_{\tau/a_k}O_k(g)\big), \qquad \langle \Omega, \psi(f)\rangle = \lim_k \omega_k(O_k(f)),$$

over the k with $\tau/a_k \in \mathbb{Z}$. It is *nontrivial* if $\mathcal{H} \neq \mathbb{C}\Omega$.

The Osterwalder–Schrader reconstruction produces such a limit from convergent renormalised Schwinger functions satisfying the axioms [26, 27]; the definition records only what is used below.

Proof of Theorem B. Let ω_k be torus limits at β_k along the tori on which $\mathrm{BP}(n_k, \varpi)$ holds. By Corollary 6.4, $\mathbb{T}_k$ has gap at least $16\varpi/(625e^2 n_k)$ on $\Omega_k^{\perp}$. For $f \in \mathcal{T}$ put $O = O_k(f)$ and $\varphi_k = \Psi_O - \omega_k(O)\Omega_k$. For dyadic τ and k large, $t = \tau/a_k$ is an integer and $16\varpi t/(625e^2 n_k) \geq \Delta\tau$ since $n_k a_k \leq \ell$, so

$$\omega_k\big(\vartheta(\bar{O})\tau_t O\big) - |\omega_k(O)|^2 = \langle \varphi_k, \mathbb{T}_k^t \varphi_k\rangle \leq e^{-\Delta\tau}\|\varphi_k\|^2.$$

Both sides converge by Definition 6.5: with $\varphi = \psi(f) - \langle\Omega, \psi(f)\rangle\Omega$, $\langle \varphi, e^{-\tau H}\varphi\rangle \leq e^{-\Delta\tau}\|\varphi\|^2$. These vectors form a subspace dense in $\Omega^{\perp}$, and the inequality extends to $\Omega^{\perp}$ and to all $\tau \geq 0$ by continuity. For $\varphi \in \Omega^{\perp}$ with spectral measure μ_φ of H this says $\int e^{-\tau E} d\mu_\varphi(E) \leq e^{-\Delta\tau}\mu_\varphi([0,\infty))$ for all τ, which forces $\mu_\varphi([0, \Delta - \eta]) = 0$ for every $\eta > 0$. So 0 is a simple eigenvalue and the spectrum on $\Omega^{\perp}$ lies in $[\Delta, \infty)$, and is not empty if $\Omega^{\perp} \neq \{0\}$. □

The proof uses the cube condition only through the decay of two-point functions of gauge-invariant observables, and the constant Δ depends on ϖ and ℓ alone.

7. The cube condition at strong coupling

Keep the setting of Sec. 6, with a bi-invariant metric δ of diameter D on G. For $V \subset E$ and $j \notin V$ let

$$C^0_{V,j} = \sup \frac{W_1^{d_V}\big(\gamma^0_V(\cdot \mid U), \gamma^0_V(\cdot \mid U')\big)}{\delta(U_j, U'_j)}, \qquad d_V(\sigma, \sigma') = \sum_{e \in V} \delta(\sigma_e, \sigma'_e),$$

the supremum over configurations that differ only at j, write $C^0_{ee'} = C^0_{\{e\},e'}$, and put $K^0_\nu = \sum_j C^0_{V_0,j}$, the sum over the links $j \notin V_0$ of direction ν. Two distinct links are partners if they share a plaquette; every link has eighteen partners, three in each of its six plaquettes.

Proposition 7.1. (i) *If $K^0_\nu \leq \alpha n^4$ for $\nu = 0, \dots, 3$ with $\alpha < 1$, then $\mathrm{BP}(n, 1-\alpha)$ holds, for every real $f \in L^2(\mu)$.*
(ii) *At $\beta = 0$, $\mathrm{BP}(n, 1)$ holds for every n.*

Proof. (i) Let $d_E(U, U') = \sum_e \delta(U_e, U'_e)$ and $\kappa_0 = (1-\alpha)n^4/L^4$. Let U, U' differ only at a link i and couple $\mathcal{K}(U, \cdot)$ and $\mathcal{K}(U', \cdot)$ by choosing the same x. If $i \in V_x$ the exteriors agree and the identity coupling costs nothing; if $i \notin V_x$ an optimal coupling of the cube laws costs at most $C^0_{V_x,i}\delta(U_i, U'_i)$, and the link i adds $\delta(U_i, U'_i)$. By translation invariance $\sum_{x:\, i \notin V_x} C^0_{V_x,i} = K^0_\nu$, so

$$W_1^{d_E}\big(\mathcal{K}(U,\cdot), \mathcal{K}(U',\cdot)\big) \leq \frac{L^4 - n^4 + \alpha n^4}{L^4}\,\delta(U_i, U'_i) = (1-\kappa_0)\delta(U_i, U'_i),$$

and by the triangle inequality $\mathcal{K}$ contracts the Lipschitz constant for d_E by $1 - \kappa_0$. Hence $\mathrm{Var}_\mu(\mathcal{K}^t h) \leq \frac{1}{2}(1-\kappa_0)^{2t}\mathrm{Lip}(h)^2 D^2 |E|^2$, while $\mathrm{Var}_\mu(\mathcal{K}^t h) = \int \lambda^{2t} d\nu_h$ for the spectral measure ν_h of $\mathcal{K}$ at $h - \mu(h)$; so ν_h lives on $[0, 1-\kappa_0]$. Lipschitz functions are dense in $L^2(\mu)$, so $\langle f, (I - \mathcal{K})f\rangle \geq \kappa_0 \mathrm{Var}_\mu(f)$ for all f, which is $\mathrm{BP}(n, 1-\alpha)$.

(ii) For $\varphi \equiv 1$, μ is a product of Haar measures. Write $f = \sum_S f_S$ with f_S a function of the links of S and $\int f_S\, dU_e = 0$ for $e \in S$; the f_S are orthogonal and $\mathcal{P}_x f = \sum_{S \cap V_x = \emptyset} f_S$, so $\sum_x \mu((f - \mathcal{P}_x f)^2) = \sum_{S \neq \emptyset} \|f_S\|^2 \#\{x : S \cap V_x \neq \emptyset\} \geq n^4 \mathrm{Var}_\mu(f)$. □

Lemma 7.2. *If $C^0_{ee'} \le c$ for all partners with $18c < 1$, then $C^0_{V,j} \le (1-18c)^{-1} \sum_{e\in V} C^0_{ej}$ for every finite V and $j \notin V$.*

Proof. Let U, U' differ only at j and let π^* minimise $\sum_{e\in V} \mathbb{E}_\pi \delta(\sigma_e, \sigma'_e)$ over couplings of $\gamma^0_V(\cdot \mid U)$ and $\gamma^0_V(\cdot \mid U')$, with $\varrho(e)$ the expected distances. Resampling one link $e \in V$ by an optimal coupling of its two single-link laws gives, by minimality, $\varrho(e) \le \sum_{k\in V} C^0_{ek}\varrho(k) + C^0_{ej}\delta(U_j, U'_j)$. Summing over e and using $\sum_{e\in V} C^0_{ek} \le 18c$ gives $(1-18c)\sum_e \varrho(e) \le \delta(U_j, U'_j)\sum_{e\in V} C^0_{ej}$. □

Lemma 7.3. *For $n \ge 2$, $L \ge 2n+4$ and a link j, $\sum_{x:\, j\notin V_x} \#\{e \in V_x : e \text{ a partner of } j\} = 18n^3 - 3n^2$.*

Proof. Let b be the base point of j and ν its direction. For each $\mu \ne \nu$ the partners in the two plaquettes of the plane $\{\mu,\nu\}$ have base points $b+\Delta$, $\Delta \in \{e_\nu, e_\mu, 0, -e_\mu, -e_\mu, e_\nu - e_\mu\}$. With $z = b - x$ and $Q = [0,n)^4$, a link based at $b+\Delta$ lies in V_x exactly when $z + \Delta \in Q$, and $j \in V_x$ when $z \in Q$. The sum is $\sum_\Delta (n^4 - |Q \cap (Q-\Delta)|)$, which is 0 for $\Delta = 0$, n^3 for $\Delta = \pm e_\mu, e_\nu$ and $2n^3 - n^2$ for $\Delta = e_\nu - e_\mu$, that is $6n^3 - n^2$ for each of the three μ. □

Lemma 7.4. *Let G have semisimple Lie algebra, δ the geodesic distance of $\langle X, Y\rangle = -\operatorname{Tr}(XY)$, and $6N|\beta| < \kappa_G$. Then $C^0_{ee'} \le N|\beta|/(\kappa_G - 6N|\beta|)$ for all partners.*

Proof. The conditional law of U_e is $\nu \propto e^{-V} dU$ with $V = -N\beta \sum_{p\ni e} \operatorname{Re}\operatorname{Tr} U_p$; by (5.1) each term has second derivative $\operatorname{Re}\operatorname{Tr}(Y^2 W)$ with $|Y| = |X|$, so $\operatorname{Ric}_G + \operatorname{Hess} V \ge K_1 = \kappa_G - 6N|\beta|$. Lemma 5.5 with one factor and the argument of Lemma 5.6 give $|\operatorname{Cov}_\nu(g,h)| \le \|\nabla g\|_\infty \|\nabla h\|_\infty / K_1$. Move the partner e' along a unit-speed geodesic; the derivative of $\nu_s(g)$ is $-\operatorname{Cov}_{\nu_s}(g, \partial_s V_s)$, and $\partial_s V_s$ comes from the one plaquette containing e and e', with gradient in U_e of norm at most $N|\beta|$ by (5.3). For smooth 1-Lipschitz g this gives $|\nu_0(g) - \nu_\delta(g)| \le N|\beta|\delta/K_1$; convolution on G preserves Lipschitz constants for a bi-invariant metric, and Kantorovich–Rubinstein duality concludes. □

Theorem 7.5. *Let $n \ge 2$ and $L \ge 2n+4$. If $C^0_{ee'} \le c$ for all partners with $18c < 1$, then $\mathrm{BP}(n, 1-\alpha_n)$ holds whenever*

$$\alpha_n = \frac{c\,(18n-3)}{(1-18c)\,n^2} < 1.$$

In particular:

(i) *for $G = \mathrm{SU}(2)$ and $|\beta_W| < \frac{2}{9}$, with $\alpha_n = |\beta_W|(18n-3)/((4 - 18|\beta_W|)n^2)$;*
(ii) *for $G \subset \mathrm{U}(N)$ with semisimple Lie algebra and $24N|\beta| < \kappa_G$, with $\alpha_n = N|\beta|(18n-3)/((\kappa_G - 24N|\beta|)n^2)$.*

For fixed β in these ranges α_n decreases to 0, so $\mathrm{BP}(n, \varpi)$ holds with $\varpi \to 1$ as $n \to \infty$, uniformly in the torus.

Proof. By translation invariance $K^0_\nu = \sum_{x:\, j\notin V_x} C^0_{V_x, j}$ for a link j of direction ν. Lemmas 7.2 and 7.3 give $K^0_\nu \le c(18n^3 - 3n^2)/(1-18c) = \alpha_n n^4$, and Proposition 7.1(i) applies. For SU(2) the single-link law is the von Mises–Fisher law and the sharp quarter gives $c = |\beta| = |\beta_W|/4$, as in Sec. 4; in general Lemma 7.4 gives $c = N|\beta|/(\kappa_G - 6N|\beta|)$, and $18c < 1$ reads $24N|\beta| < \kappa_G$. □

For SU(2) the condition $\alpha_n < 1$ reads $|\beta_W|(18n^2 + 18n - 3) < 4n^2$; it holds for $|\beta_W| < \frac{16}{105}$ at $n = 2$, and its threshold increases to $\frac{2}{9}$ as $n \to \infty$. At strong coupling the condition is therefore *easier* for large cubes, because the response of a cube to one boundary link is spread over n^4 links while it touches only $O(n^3)$ of them.

8. Free massive fields

Here $d \ge 1$, $n \ge 2$, $L \ge n+1$, $\mathbb{T} = (\mathbb{Z}/L\mathbb{Z})^d$, $\mu > 0$ is a lattice mass, $-\Delta$ is the lattice Laplacian and $Q = -\Delta + \mu^2$. For a positive definite Q on $\mathbb{R}^N$ let γ_Q be the centred Gaussian

law with covariance $C = Q^{-1}$, and for $V \subset \{1, \dots, N\}$ let $\mathcal{P}_V$ be the conditional expectation given the coordinates outside V. For the cubes $V_x = x + \{0, \dots, n-1\}^d$, $x \in \mathbb{T}$, put

$$\varpi(Q) = \inf_f \frac{n^{-d} \sum_x \|f - \mathcal{P}_{V_x} f\|^2}{\mathrm{Var}_{\gamma_Q}(f)}, \tag{8.1}$$

the infimum over nonconstant $f \in L^2(\gamma_Q)$, the best constant in $\mathrm{BP}(n, \cdot)$. With $s = n\mu$ and $\mu = ma$, $n = \ell/a$, the parameter $s = m\ell$ is the mass times the physical size of the cube.

Lemma 8.1 (linear functions suffice)**.** *For a positive definite Q and any family of coordinate sets (V_x), the infimum in (8.1) equals its infimum over the linear functions $\ell_h(\phi) = \langle h, \phi \rangle$, and*

$$\|\ell_h - \mathcal{P}_V \ell_h\|^2 = \langle h_V, Q_{VV}^{-1} h_V \rangle, \qquad \mathrm{Var}(\ell_h) = \langle h, Q^{-1} h \rangle,$$

with h_V the restriction of h to V and Q_{VV} the block of Q on V.

Proof. The conditional law of ϕ_V given ϕ_{V^c} is Gaussian with covariance Q_{VV}^{-1}, which gives the formula for ℓ_h. Write $\phi = Q^{-1/2}\xi$ with ξ standard Gaussian. Conditioning on ϕ_{V^c} is conditioning on the Gaussian subspace $K_V = \{\langle u, \xi \rangle : u \in Q^{-1/2}\mathbb{R}^{V^c}\}$, and on the m-th Wiener chaos, identified with the symmetric tensors $\mathrm{Sym}^m(\mathbb{R}^N)$, it acts as $\pi_V^{\otimes m}$, with π_V the orthogonal projection onto $Q^{-1/2}\mathbb{R}^{V^c}$ [20, Ch. 4]. The chaoses are orthogonal and invariant under every $\mathcal{P}_V$, so it suffices to treat one chaos $m \geq 1$. The projection $I - \pi_V^{\otimes m}$ dominates $(I - \pi_V) \otimes I^{\otimes(m-1)}$, as both are projections that commute and the range of the second lies in that of the first. For a symmetric tensor F, therefore,

$$\sum_x \|F - \pi_{V_x}^{\otimes m} F\|^2 \geq \Big\langle F, \Big(\sum_x (I - \pi_{V_x})\Big) \otimes I^{\otimes(m-1)} F \Big\rangle \geq \lambda \|F\|^2,$$

where λ is the least eigenvalue of $\sum_x (I - \pi_{V_x})$ on $\mathbb{R}^N$, which is the infimum of the quotient over linear functions ($m = 1$). □

Theorem 8.2 (the exact constant)**.** *Let $V = \{0, \dots, n-1\}^d$, $-\Delta_D$ its Dirichlet Laplacian and $\mathbf{1}$ its constant function. Then*

$$\varpi(-\Delta + \mu^2) = \frac{\mu^2}{n^d} \langle \mathbf{1}, (-\Delta_D + \mu^2)^{-1} \mathbf{1} \rangle,$$

attained only at multiples of $\ell_{\mathbf{1}}$ among linear functions, and

$$\Big(\frac{5}{6}\Big)^d \frac{s^2}{s^2 + 10d} \leq \varpi(-\Delta + \mu^2) \leq \frac{s^2}{s^2 + 4dn^2 \sin^2 \frac{\pi}{2n+2}} \leq \frac{s^2}{s^2 + 16d/9}.$$

Proof. Since $L \geq n+1$, the block $Q_{V_x V_x}$ is the translate of $Q_{VV} = -\Delta_D + \mu^2$; put $G_V = Q_{VV}^{-1}$. By Lemma 8.1, $n^d \varpi$ is the least value of $\langle h, Sh \rangle / \langle h, Q^{-1} h \rangle$ with $S = \sum_x R_{V_x}^{\mathsf{T}} Q_{V_x V_x}^{-1} R_{V_x}$, that is, the least eigenvalue of $Q^{1/2} S Q^{1/2}$. Translations commute with Q and S, so the plane waves $e_k(y) = e^{ik \cdot y}$ diagonalise it with eigenvalues $\lambda_k = \hat{q}(k) \langle e_k^V, G_V e_k^V \rangle$, where $\hat{q}(k) = \mu^2 + \sum_i (2 - 2\cos k_i)$ and e_k^V is the restriction to V.

Let B map f to the function $u \mapsto \sum_{w \sim u,\, w \notin V} f(w)$ on V. Then $Q_{VV} R_V f = R_V Q f + B f$, so $Q_{VV} e_k^V = \hat{q}(k) e_k^V + B e_k$, and pairing G_V of this with e_k^V gives

$$\lambda_k = n^d - \sum_{u, u' \in V} \sum_{w \sim u',\, w \notin V} G_V(u, u') \cos(k \cdot (w - u)).$$

Write $Q_{VV} = (2d + \mu^2)(I - K)$ with K entrywise nonnegative and $\|K\| \leq 2d/(2d + \mu^2) < 1$; the Neumann series shows that G_V has positive entries. Hence $\lambda_k - \lambda_0 = \sum G_V(u, u')(1 - \cos(k \cdot (w - u))) \geq 0$, with equality only at $k = 0$ (take $u = u'$ on a face and $w = u + e_i$). Since $Q\mathbf{1} = \mu^2 \mathbf{1}$, $\lambda_0 = \mu^2 \langle \mathbf{1}, G_V \mathbf{1} \rangle$, which is the formula.

Upper bound. $G_V \leq (\lambda_1 + \mu^2)^{-1}$ with $\lambda_1 = 4d \sin^2 \frac{\pi}{2n+2}$ the least eigenvalue of $-\Delta_D$, and $\|\mathbf{1}\|^2 = n^d$; then $\sin t \geq 2t/\pi$ gives $n^2 \lambda_1 \geq 4dn^2/(n+1)^2 \geq 16d/9$.

Lower bound. By the variational formula $\langle \mathbf{1}, G_V \mathbf{1}\rangle \geq \langle \mathbf{1}, w\rangle^2/\langle w, Q_{VV} w\rangle$, take $w(u) = \prod_i \tau(u_i)$ with the parabolic profile $\tau(s) = 6(s+1)(n-s)/(n(n+1)(n+2))$ on $\{0,\dots,n-1\}$, zero outside. Then $\sum \tau = 1$, $S_2 = \sum \tau^2 = \alpha_n/n$ and $D_2 = \sum_{s\in\mathbb{Z}} (\tau(s+1)-\tau(s))^2 = \beta_n/n^3$ with

$$\alpha_n = \frac{6(n^2+2n+2)}{5(n+1)(n+2)} \leq \frac{6}{5}, \qquad \beta_n = \frac{12n^2}{(n+1)(n+2)} = 10\rho_n\alpha_n, \quad \rho_n = \frac{n^2}{n^2+2n+2} < 1,$$

by the power sums $\sum_{u=1}^n u(n+1-u) = n(n+1)(n+2)/6$, $\sum_{u=1}^n u^2(n+1-u)^2 = n(n+1)(n+2)(n^2+2n+2)/30$ and $\sum_{u=0}^n (n-2u)^2 = n(n+1)(n+2)/3$. The Dirichlet form of the product is $\langle w, -\Delta_D w\rangle = dS_2^{d-1} D_2$ and $\langle \mathbf{1}, w\rangle = 1$, so

$$\varpi \geq \frac{\mu^2}{n^d(\mu^2 S_2^d + dS_2^{d-1} D_2)} = \frac{1}{\alpha_n^d + d\alpha_n^{d-1}\beta_n s^{-2}} = \alpha_n^{-d}\frac{s^2}{s^2+10d\rho_n} \geq \Big(\frac{5}{6}\Big)^d \frac{s^2}{s^2+10d}. \quad \square$$

At $n=2$ the two bounds coincide with $s^2/(s^2+4d)$. For $d=4$ the lower bound is 0.0118 at $s=1$ and 0.0438 at $s=2$; the upper bound vanishes as $s\to 0$, so a massless free field satisfies no cube condition uniform in the lattice spacing. The same proof, with the Hodge identity of the torus, gives the cube condition for the free massive vector (Proca) field on gauge-invariant functions, with the same lower bound and the upper bound $s^2/(1+s^2)$ [4].

9. Weak coupling: what a proof of the mass gap still needs

The continuum limit of four-dimensional Yang–Mills theory is taken along the tuning dictated by asymptotic freedom: for SU(N) the bare coupling $g^2 = 2N/\beta_W$ tends to zero like $1/(b_0 \log(1/(a\Lambda)))$ with $b_0 = 11N/(48\pi^2)$ and a fixed physical scale Λ, so $\beta_W \to \infty$ logarithmically in $1/a$. Theorems A and 7.5 hold for $\beta_W < N^2/29$ and $\beta_W < N^2/48$ respectively; at fixed β the lattice gap in physical units, $m_G(\beta)/a$, diverges as $a\to 0$, and no continuum limit exists at fixed β. The results of this paper therefore say nothing about the continuum theory without statement (D). Three facts show what (D) asks for.

(a) A cube condition is a mass. By Theorem 8.2 the best constant for a free field is at most $s^2/(s^2+16d/9)$ with $s = m\ell$. At weak coupling the leading approximation of the Wilson measure, in any gauge, is a massless Gaussian field, for which the constant vanishes as $a\to 0$ at fixed ℓ. Hence (D) does not follow from perturbation theory to any finite order: it holds only if the interaction generates a mass of order $1/\ell$, and it is a quantitative form of that mass generation.

(b) The cube must grow. For SU(N), $\beta \geq 1$ and the torus of side $2n+4$, BP(n,ϖ) implies $\varpi \leq C_N n^4(1+\log\beta)/\beta$ [4]: on a small torus at weak coupling the Polyakov loops order, and the cube dynamics can change their centre charge only through the curvature, which is small. Along the tuning $n_k = \ell/a_k$ grows like a power of $1/a_k$ while β_k grows like $\log(1/a_k)$, so this obstruction is consistent with (D) but excludes every argument at fixed cube size.

(c) The group must be non-abelian. For $G = $ U(1) with the Villain action at large β the theory has a massless photon [17], and the cube condition fails at every cube size, since by Theorem 6.3 it would make the field-strength correlations summable [4]. Every proof of (D) must therefore use the non-abelian structure of G, exactly as a proof of the mass gap must.

The two ends. Bałaban's renormalisation group [7, 8, 9, 13] controls the effective actions of the lattice theory from the scale a up to scales at which the running coupling is still small, uniformly in the lattice spacing, in finite volume. The theorems of Secs. 5–7 control the theory when the coupling is large: they need $N\beta$ small, that is, g large. The running coupling grows towards the infrared, so a proof of (D) would follow the renormalisation group from the weak-coupling regime, where Bałaban's estimates hold, through the regime of coupling of order one, to a scale ℓ at which the effective theory satisfies a strong-coupling criterion of the kind proved here, and would then transport the resulting cube estimate back to the fine lattice. Neither the middle step nor the final transport is available: the effective actions are not of Wilson form, Bałaban's analysis stops before the coupling is

of order one, and no known criterion accepts the effective theory at that scale. This gap between the two ends is the Yang–Mills mass gap problem in the form isolated here.

Statement (E) is the construction problem. Bałaban's ultraviolet stability bounds give compactness of suitably smeared observables in finite volume, but the infinite-volume limit, Euclidean invariance and nontriviality of the limit are open [19]. Given (E), Theorem B reduces the mass gap to (D) with the explicit constant $\Delta \geq 16\varpi/(625e^2\ell)$.

10. Conclusion

Lattice Yang–Mills theory in four dimensions has a mass gap with the explicit lower bound $m_G(\beta)$ for every compact simple gauge group in the range $N\beta < \frac{2}{29}\kappa_G$, and for SU(2) up to $\beta_W = \frac{1}{3}$. A single spectral gap ϖ of the dynamics that resamples cubes of a fixed physical size ℓ, uniform along the continuum tuning, gives every nontrivial continuum limit the mass gap $16\varpi/(625e^2\ell)$. That spectral gap holds at strong coupling and for massive free fields, it fails for massless fields, at fixed cube size and in the abelian Coulomb phase, and along the weak-coupling tuning required by the continuum problem it is not proved. The Clay problem remains open; what this paper contributes is an explicit strong-coupling theory for every gauge group and a reduction of the continuum gap to one estimate on one scale.

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Declarations

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Appendix A. The line field

Table 2 lists, for each residue class of x modulo two, the direction $\sigma(x)$ and the values $\pi_x(\mu) = \sigma(x + e_\mu)$. In every row π_x fixes $\sigma(x)$ and is a three-cycle on the other directions.

Table 2. The line field: x modulo two (digits $x_0x_1x_2x_3$), $\sigma(x)$, and $\sigma(x+e_\mu)$ for $\mu = 0, 1, 2, 3$.

x	σ	0	1	2	3	x	σ	0	1	2	3
0000	0	0	2	3	1	0001	1	2	1	3	0
1000	0	0	3	1	2	1001	2	1	3	2	0
0100	2	3	0	2	1	0101	1	3	1	0	2
1100	3	2	0	1	3	1101	3	1	2	0	3
0010	3	1	2	0	3	0011	3	2	0	1	3
1010	1	3	1	0	2	1011	2	3	0	2	1
0110	2	1	3	2	0	0111	0	0	3	1	2
1110	1	2	1	3	0	1111	0	0	2	3	1

Appendix B. The blocks of Lemma 5.2

Diagonal blocks. Let $w_a = e^{i\phi}$. On the diagonal part $f_p = -\cos\phi\,(y_1+y_2-y_3-y_4)^2$ and $c_p = 2(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$, so

$$\mathcal{F} = \sum_i \alpha_i y_i^2 + \cos\phi\,(y_1+y_2-y_3-y_4)^2 - \tfrac{9}{5}(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4),$$

with $y_k = 0$, $\alpha_{k\pm1} = \frac{21}{10}$, $\alpha_{k+2} = \frac{5}{4}$. It is affine in $\cos\phi$, so positivity at $\cos\phi = \pm1$ suffices. For $k=1$ and $\cos\phi = 1$ the matrix in (y_2, y_3, y_4) is

$$\begin{pmatrix} \frac{31}{10} & -\frac{19}{10} & -1 \\ -\frac{19}{10} & \frac{9}{4} & \frac{19}{10} \\ -1 & \frac{19}{10} & \frac{31}{10} \end{pmatrix},$$

with leading principal minors $\frac{31}{10}$, $\frac{673}{200}$, $\frac{8421}{2000}$. In general the leading principal minors are $\frac{31}{10}, \frac{673}{200}, \frac{8421}{2000}$ (fixed link e_1 or e_4), $\frac{31}{10}, \frac{861}{100}, \frac{8421}{2000}$ (e_2) and $\frac{9}{4}, \frac{673}{200}, \frac{8421}{2000}$ (e_3) at $\cos\phi = 1$, and $\frac{11}{10}, \frac{53}{200}, \frac{21}{2000}$ (e_1, e_4), $\frac{11}{10}, \frac{21}{100}, \frac{21}{2000}$ (e_2) and $\frac{1}{4}, \frac{53}{200}, \frac{21}{2000}$ (e_3) at $\cos\phi = -1$; all are positive.

Pair blocks. Let $w_a = e^{i(\chi+\vartheta)}$ and $w_b = e^{i(\chi-\vartheta)}$. The terms of f_p carry $w_a + \bar{w}_b = 2\cos\chi\, e^{i\vartheta}$ or its conjugate, and c_p carries $w_a\bar{w}_b = e^{2i\vartheta}$, so the Hermitian matrix $M(m,\vartheta)$ of $\mathcal{F}$, $m = \cos\chi$, is affine in m with $M(-1,\vartheta) = M(1,\vartheta+\pi)$, and $m = 1$ suffices. For the fixed link e_2, in the variables (z_1, z_3, z_4) and with $c = \cos\vartheta$,

$$M = \begin{pmatrix} \frac{21}{5} + 2c & -2e^{i\vartheta} & -\frac{9}{5} - 2e^{i\vartheta} \\ -2e^{-i\vartheta} & \frac{21}{5} + 2c & \frac{9}{5} + 2e^{-i\vartheta} \\ -\frac{9}{5} - 2e^{-i\vartheta} & \frac{9}{5} + 2e^{i\vartheta} & \frac{5}{2} + 2c \end{pmatrix}.$$

With $u = \cos^2(\vartheta/2) \in [0,1]$ its leading principal minors are

$$\tfrac{20u+11}{5}, \qquad \tfrac{(20u+1)(20u+21)}{25}, \qquad \tfrac{400u(40u^2-23u+4)+21}{250},$$

and $40u^2 - 23u + 4$ has negative discriminant $529 - 640$. The other positions give the same determinant and the leading minors $\frac{20u+11}{5}$ or $\frac{8u+1}{2}$ and $\frac{800u^2-180u+53}{50}$, whose discriminant $32400 - 169600$ is negative. Every block is positive definite.

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